

2-2 $\vec{v} \rightarrow \boxed{A} \boxed{B}$
 m_1, m_2

$\Delta t_1, \Delta t_2, \vec{F}$

$\vec{F} \Delta t_1 = (m_1 + m_2) V_1 - 0$ ①

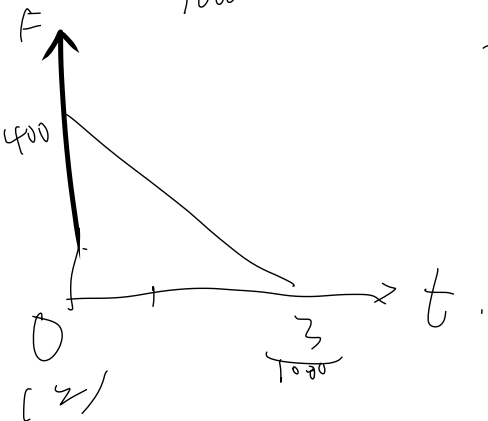
$\vec{F} \Delta t_2 = m_2 V_2 - m_2 V_1$ ②

$\Rightarrow V_1 = \frac{\vec{F} \Delta t_1}{m_1 + m_2}$

$V_2 = \frac{\vec{F} \Delta t_2 + m_2 \vec{F} \Delta t_1}{m_2}$

2-4, 300 m/s
 $\vec{F} = 400 - \frac{4 \times 10^5}{3} t$

(1) $t = \frac{3}{1000} \text{ s}$ 时, $\vec{F} = 0$



$-\frac{400000}{3}$

$\vec{I} = \int_0^{t_1} (400 - \frac{4 \times 10^5}{3} t) dt$

$\vec{I} = 400t - \frac{4 \times 10^5}{6} t^2 \Big|_0^{\frac{3}{1000}} = \frac{3}{5} \text{ N} \cdot \text{s}$

$\frac{4 \times 10^5}{3} = \frac{4 \times 10^5}{3} t$

$\frac{6}{5} - \frac{3}{10} = \frac{12}{10} - \frac{3}{10} = \frac{9}{10}$

$\frac{3}{500}$

$\vec{F} = 400 - \frac{4 \times 10^5}{3} t_1 = 0$ ①

$\int_0^{t_1} \vec{F} dt_1 = \vec{I}$ ②

$\vec{I} = 300 \text{ N} - 0$ ③

$\Rightarrow \begin{cases} t_1 = \frac{3}{1000} \\ m = \frac{1}{500} \text{ kg} \end{cases}$

2-14

$\vec{v} \rightarrow \boxed{m} \boxed{m}$
 m_0, m
 $\text{相对车 } V_{\text{车}} = u$

(1) $m_0 V_0 = 2mu$ ①

$\Rightarrow V_0 = \frac{2mu}{m_0}$

(2) $(m + m_0) V_1 = mu$ ① $m(V_1 - u)$

$(m + m_0) V_1 = m_0 V_2 + mu$ ②

$\Rightarrow V_1 = \frac{mu}{m + m_0}$

$V_2 = \frac{(m + m_0) V_1 + mu}{m_0}$
 $= \frac{mu}{m_0} + \frac{mu}{m_0 + m}$

(3)

$V_0 = \frac{2mu}{m_0}$

$V_2 = \frac{mu}{m_0} + \frac{mu}{m_0 + m}$

$\frac{mu}{m_0} - \frac{mu}{m_0 + m}$

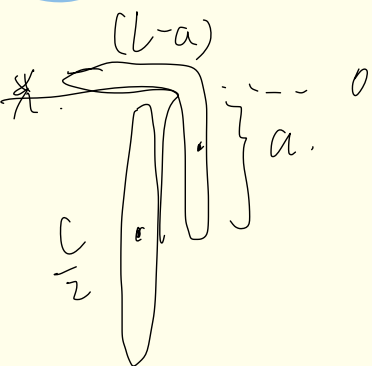
$\frac{(m_0 + m)mu - mu m_0}{m_0(m_0 + m)}$

$mu(m_0 + m - m_0)$

$V_0 - V_2 = \frac{2mu}{m_0(m_0 + m)} > 0$

$V_0 > V_2$

2-20.

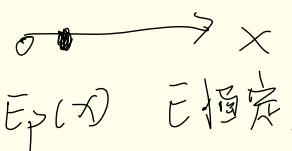


$$-\left(\frac{a}{L}mg\right)\frac{L}{2} = -mg\frac{L}{2} + \frac{1}{2}mV^2 \quad (1)$$

$$-\frac{a^2}{2L}mg = -\frac{mgL}{2} + \frac{1}{2}mV^2$$

$$\Rightarrow V = \sqrt{\frac{g(L^2 - a^2)}{2L}} = \frac{V}{4}$$

2-27



$$t = \int_0^x \frac{dx}{\sqrt{\frac{2}{m}(\bar{E} - \bar{E}_p(x))}}$$

$$dx = v dt$$

$$dt = \frac{dx}{v}$$

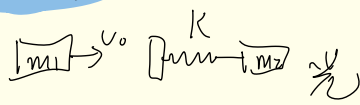
$$\bar{E} = \bar{E}_p(x) + \bar{E}_K = \bar{E}_p(x) + \frac{1}{2}mV^2$$

$$\bar{E}_K = \frac{mV^2}{2} \quad V^2 = \frac{2\bar{E}_K}{m} = \frac{2}{m}(\bar{E} - \bar{E}_p(x))$$

$$V = \sqrt{\frac{2}{m}(\bar{E} - \bar{E}_p(x))}$$

$$\therefore \int_0^t dt = \int_0^x \frac{dx}{v} = \int_0^x \frac{dx}{\sqrt{\frac{2}{m}(\bar{E} - \bar{E}_p(x))}} = t$$

2-36.



$$F_{12} \leftarrow m_1 m_2 \rightarrow F_{21}$$

$$m_1 v_0 = (m_1 + m_2) v \quad (1)$$

$$\frac{1}{2}m_1 v_0^2 = \frac{1}{2}(m_1 + m_2) v^2 + W_K \quad (2)$$

$$W_K = \frac{Kx^2}{2} \quad (3)$$

$$\Rightarrow x = v_0 \sqrt{\frac{m_1 m_2}{K(m_1 + m_2)}}$$

2-53.



$$\frac{Gmm_0}{R}$$

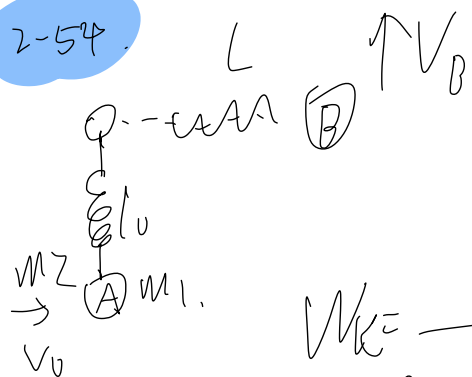
$$\frac{1}{2}m_0 v_0^2 = \frac{1}{2}m_0 v^2 = \frac{Gmm_0}{R} \quad (1)$$

$$m_0 v_0 h = m_0 v R \quad (2)$$

$$\Rightarrow v = \sqrt{v_0^2 + \frac{2GM}{R}}$$

$$h = \frac{v}{v_0} R = \frac{v}{v_0} R = \sqrt{1 + \frac{2GM}{v_0^2 R}} R$$

2-54.



$$W_K = \frac{K(L-l_0)^2}{2} \quad (3)$$

$$m_2 V_0 = (m_1 + m_2) V_{\text{共}} \quad (1)$$

$$\frac{1}{2} (m_1 + m_2) V_{\text{共}}^2 = \frac{1}{2} (m_1 + m_2) V_B^2 + W_K \quad (2)$$

$$\Rightarrow V_B = \sqrt{\frac{(m_1 V_0)^2}{(m_1 + m_2)^2} - \frac{K(L-l_0)^2}{m_1 + m_2}}$$

$$\vec{T}_1 \cdot \vec{r} = 0$$